

Minguhn Kim

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RESEARCH INTERESTS

Primary Focus: AI-native Semiconductor Devices, Neuromorphic Computing, Cross-layer Optimization (Device-Circuit-Algorithm co-design).

Technical Interests: Machine Learning-based Device Modeling, Physical Mechanism Exploration, Next-generation Logic/Memory Devices.

EDUCATION

POSTECH (Pohang University of Science and Technology) Feb. 2025 – Feb. 2027 (Expected)
M.S. in Graduate School of Semiconductor Technology Pohang, South Korea

- Research Focus: AI-driven RL-Inspired Circuit Optimization, AI-based Analog Circuit Layout Automation

Chung-Ang University (CAU) Mar. 2017 – Feb. 2024
B.S. in Electrical and Electronic Engineering Seoul, South Korea

- **GPA: 4.34 / 4.50 (Summa Cum Laude)**
- Military Leave of Absence: 2 years of mandatory military service in the ROK Army 702nd Special Commando Battalion

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, NumPy, Pandas), C/C++ (Basic).
- **Design Tools:** Cadence Virtuoso (Custom IC Design/Layout), VS Code (Development/Debugging), Git (Cooperation).
- **Algorithms:** Reinforcement Learning (Actor-Critic), A* Search Algorithm.

RESEARCH EXPERIENCE

POSTECH Broadband Electronics by Virtual Intelligence Laboratory (BEVIL) Graduate Student

Project: CTLE Optimization using Reinforcement Learning 2025 – Present

- Developed a Python-based framework to optimize Continuous-Time Linear Equalizer (CTLE) parameters using Actor-Critic algorithms.
- Bridged the gap between analog circuit characteristics and AI-driven automated design flows.

Samsung AI Cluster Project Lead Developer

Project: DDI Layout Automation via A Search Algorithm* 2025 – 2026

- Implemented a routing automation GUI tool for Display Driver IC (DDI) layout designs.
- Managed complex function flows and software debugging to enhance design efficiency.

AWARDS & LANGUAGES

- **OPIc (English Proficiency):** Advanced Low (AL) Jan. 2026
- **Academic Excellence Scholarship:** Chung-Ang University (Multiple semesters)

EXTRACURRICULAR ACTIVITIES

- **Music Production:** Skilled in digital audio workstations (DAWs) and piano performance; focuses on creative problem-solving and meticulous attention to detail.